**Lab 2- Data Structure**

**Q 1)**

**Topic:** Data Structures — Stack, Arrays, Classes
**Objective:** Implement a fixed-size stack of characters and use it to check whether parentheses in an expression are balanced.

Write a C++ program that implements a **Stack class using a dynamically allocated array of characters (char)**.

The stack must have a fixed maximum capacity specified when the object is created.

The program must use this stack to determine whether parentheses () in a mathematical expression are balanced.

Q 2)

**Project Level:** Intermediate – Advanced
**Programming Language:** C++
**Topics:** Arrays, Sorting Algorithms, Functions, Recursion, Random Numbers, Time Measurement, and Performance Analysis.

Develop a C++ program to compare the execution time of five sorting algorithms:

1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort

The program must generate **five large arrays containing random integers between 1 and 100**, execute each sorting algorithm on identical input data, measure the execution time, and calculate the average execution time for each algorithm.

The objective is to determine which sorting algorithm performs best when processing large datasets.

Q 3)

**C++ Programming Project: Bank Queue Management Simulation**

**Project Level:** Intermediate – Advanced
**Programming Language:** C++
**Topics:** Data Structures, Dynamic Arrays, Queues, Classes, Simulation, Random Numbers, and Performance Analysis.

**1. Project Objective**

Design and implement a **Bank Queue Management Simulation System** using C++ and Object-Oriented Programming (OOP).

The system must simulate customers arriving at a bank and waiting to receive service.

Each service counter has its own queue, implemented using a **dynamically allocated array**.

The simulation must determine how many service counters (queues) the bank needs based on:

- Customer arrival rate.
- Customer service rate.
- Number of waiting customers.
- Maximum allowed queue length.
- Total simulation time.

**Main Business Rule**

**If the number of waiting customers in a queue exceeds 5, the bank must open a new service counter with its own queue.**

For the simulation, assume that the customer causing the queue to exceed five is assigned to the newly opened queue, so the original queue does not continue growing.

**2. Project Scenario**

A bank initially operates with **one service counter**.

Customers arrive at random intervals and join a waiting queue.

Each service counter serves one customer at a time.

When the waiting queue becomes too long, the bank automatically opens another counter.